Akash Rao

GRADUATE RESEARCH ASSISTANT, ICAMS & PHD STUDENT, COMPUTER SCIENCE

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Education

Auburn University, Auburn, AL

PhD in Computer Science (January 2025 - Present) | GPA – 3.8/4.0

Auburn University, Auburn, AL

MS Data Science and Engineering (August 2023 - August 2025) | GPA – 3.8/4.0

AMC Engineering College, Bengaluru, India

BTech Computer Science (August 2017 - August 2021) | GPA – 3.7/4.0

Publications

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1. *"FSP-DETR: Few-shot Prototypical Parasitic Ova Detection"* ,Shubham Trehan, Udhav Ramachandran, **Akash Rao**, Ruth Scimeca and S.N Aakur, https://openaccess.thecvf.com/content/WACV2026/html/Trehan_FSP-DETR_Few-Shot_Prototypical_Parasitic_Ova_Detection_WACV_2026_paper.html(Co-First Author, Accepted WACV 2026)
    - Led the design, implementation, and evaluation of transformer-based models for microscopic parasitic ova detection utilizing pytorch.
    - Processed and evaluated models on a novel, custom-annotated dataset alongside standard BCCD and malaria-infection datasets.
    - Investigated the effects of varying sample sizes () for few-shot learning, establishing robust performance baselines for data-scarce medical imaging environments.

2. *"HYPERTAXA: Zero-Shot Microbial Taxonomic Inference Framework"* , **Akash Rao**, Sushim Gupta, Vishalini Ramnath, Akhilesh Ramachandran, and S N. Aakur, (Under Review, Scientific Reports)
    - Developed a novel zero-shot (ZSL) and generalized zero-shot (GZSL) deep learning framework using tensorflow to identify previously unseen bacterial species, addressing the limitation of supervised models when reference databases are incomplete.
    - Engineered a multiscale hyperspectral k-mer transition field encoding that transforms raw genome sequences into structured multichannel tensors, capturing both local sequence composition and long-range genomic patterns.
    - Implemented a two-stage node2vec MDS embedding to align genomic representations to a taxonomy-aware semantic space, preserving hierarchical genus-family-order relationships.
    - Achieved state-of-the-art performance, outperforming transformer-graph and pseudo-image baselines on the TEPI benchmark and a 1,110-species large-scale evaluation.
    - Demonstrated practical utility in clinical settings via a few-shot adaptation to a 100-pathogen panel, enabling rapid pathogen discovery and BLAST-restricted confirmation on raw sequencing data.

Projects

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1. Prototypical Networks and Latent Space Optimization (Current Research)
    - Ongoing research on prototypical networks, specifically focusing on shaping the latent space to enhance few-shot learning performance and feature clustering.

2. Advance Video Analysis via DETR
    - Engineered a robust video analysis and object detection pipeline utilizing Detection Transformers (DETR) and Python.
    - Processed and analyzed complex video datasets, applying transformer-based architectures to extract spatial-temporal features

Work Experience

Graduate Research Assistant, ICAMS

October 2024 – Present

- Led the design and implementation of custom operational software using Python and React, collaborating with Industrial Engineering teams to translate complex requirements into scalable full-stack solutions.
- Built a camera surveillance application featuring motion detection and real-time video prioritization, leveraging React, MQTT, Docker, and Flask, deployed on a Raspberry Pi.
- Developed a specialized job management application tailored for lab environments to streamline task assignment, tracking, and operational efficiency.
- Managed and mentored a team of undergraduate developers, ensuring code quality and timely delivery of project milestones.

Software Engineer, Dcode Solutions L.L.C

October 2021 – June 2023

- Developed and deployed integrated Android applications using **Java** and **MSSQL Server**, aligning product features with client business process requirements.
- Designed and launched an Asset Tracking System for the Ministry of Monitor and Control, U.A.E. government, resulting in improved asset visibility.
- Led the end-to-end development of a warehouse management system for Dubai Duty-Free, utilizing Bartender Software for automated barcode tracking of inventory items.
- Conducted client training sessions and provided technical documentation for custom software deployments, supporting pre-sales and adoption phases.
- Analyzed and interpreted large datasets with advanced analytical tools, providing actionable insights that enhanced operational efficiency.

Software Engineer Intern, Scientia Innovations

October 2020 – December 2020

- Built a cross-platform e-commerce and educational course discovery application using Flutter.

Skills and Certifications

Programming languages

- Python
- React
- SQL
- Java

Machine Learning Libraries

- Tensorflow
- Pytorch
- Scikit-learn
- Gensim

Certifications

- Machine Learning, Stanford University, Coursera
- Industry 4.0
- Docker for Developers

Mobile Application Development

- React Native
- Flutter
- Native Android